

ESSIE CHENG

(812) 219-9960 | e2ssiecheng@gmail.com | www.linkedin.com/in/essiecheng | github.com/essiecheng

EDUCATION

University of California, San Diego, Halicioglu Data Science Institute
Bachelor of Science in Data Science

Sep 2022 - June 2026

Relevant Coursework: DS in Practice, Machine Learning, Data Structures and Algorithms, Data Management, Data Visualization, Scalable Analytics, Statistical Methods, Product Marketing and Management, Business Analytics

Skills/Tools: Python, SQL, JavaScript, HTML/CSS, Tableau, Power BI, Git, NumPy, Pandas, Matplotlib, Seaborn, Plotly, SciKit-Learn, Keras, TensorFlow, Spark, Dask, AWS, Palantir Foundry, Microsoft Suite (Excel, Word, Powerpoint)

EXPERIENCE

Cook Medical - Data Science Intern

Jun 2025 - Aug 2025

- Automated Bill of Process generation in Palantir Foundry by building data pipelines, cleaning and transforming manufacturing data, and applying LLM-based information extraction – reducing build time from weeks to minutes
- Improved product stability of CookGPT for hundreds of internal users by designing 19 Azure test cases, executing QA sprints, and validating data integrity through Cosmos DB, uncovering and documenting high-priority bugs

JP Morgan Chase - Break Through Tech AI Studio Fellow

Aug 2024 - Dec 2024

- Led project team to analyze impact of financial news sentiments on stock price movements through a series of data preprocessing, statistical testing, and natural language processing techniques
- Integrated financial news analysis and stock price data to leverage supervised learning models for enhanced stock price movement prediction, achieving 70% directional accuracy and driving informed investment decisions

PROJECTS

NBA Game Outcome Predictor

Dec 2024

- Engineered comprehensive feature set and conducted exploratory data analysis on 10 years of NBA game data to uncover historical matchup patterns, team performance insights, and seasonal trends
- Implemented and fine-tuned multiple classifiers (Logistic Regression, Decision Trees, Naive Bayes, Ensembling) to predict win/loss game results, achieving 80% test accuracy

Power Outages Analysis and Prediction

Nov 2023 - Dec 2023

- Analyzed U.S. major power outage data, performing hypothesis and permutation tests for missingness assessment and statistical investigations; visualized findings and identified regions for electricity infrastructure improvement
- Developed and optimized a random forest regression model to predict the severity of a major power outage in terms of its outage duration, reducing RMSE from baseline by ~400 minutes

United States in the Olympics

Feb 2024 - Mar 2024

- Collaborated to design and develop 8 interactive data visualizations using D3.js, highlighting key trends in the U.S. Olympic journey and identifying opportunities for Olympic future success
- Created a scrollytelling webpage using Svelte, integrating visualizations, narratives, and user interaction features to showcase insights and enhance data exploration

ACTIVITIES & INVOLVEMENT

Sustainable Investment Group - Quantitative Analyst

Jan 2024 - Present

- Implemented financial models to analyze and predict stock performance, supporting sustainable investment strategies

Data Science Student Society - Marketing Chair

May 2024 - Present

- Created 20+ engaging Instagram reels (3-5k views) and posts, increasing audience engagement by >100%

Alpha Phi Omega Service Fraternity - Networking Chair / Member

Sep 2024 - Feb 2025

- Fostered professional growth for 90+ members, promoted 30+ professional events, and volunteered 45+ hours

Break Through Tech AI - Fellow

May 2024 - May 2025

- Selected from 3000+ applicants to participate in the Break Through AI program by Cornell Tech, earning a Cornell University online certificate in Machine Learning Foundations, experiential learning, and industry mentorship